Learning Programme Guide

How the sessions, experiments and handouts fit together — for teachers and volunteers

WHAT THIS PACKAGE IS
A set of ready-to-use lessons, hands-on experiments and reference sheets that introduce school students to molecular biology, microbiology, ecology and synthetic biology. Each document has a suggested time, a list of learning goals and a short "check your understanding" section.

1. Document map

#	Document	Type	Time	Suggested level
01	From DNA to Protein	Talk + notes	90 min	Classes 9–12
02	Codon Lottery (+ GeneDetectives)	Game	20–40 min	After 01
03	Genetics of the Microworld (lac operon, quorum sensing)	Talk + worksheet + games	75–90 min	Classes 9–12
04	Plasmid Design and Cloning	Talk + notes	90 min	Classes 10–12
05	Bacterial Culture (Ek Pehal)	Talk	60–75 min	Classes 9–12
06	Agar Plate Art & Streaking	Hands-on	30 min + growth	Classes 9–12
07	Viral Diagnosis: PCR	Talk + exercise	75 min	Classes 11–12+
08	Enzymes	Talk + demo	45–60 min	Classes 9–12
09	Yeast Metabolic Rates	Hands-on	75 min	Classes 8–12
10	Fingerprint Development	Hands-on	45–60 min	Classes 8–12
11	Eco-Lies: Ecology	Talk + game	60–90 min	Classes 9–12
12	Biomagnification Activity	Hands-on	30–40 min	Classes 8–12
13	Synthetic Biology Primer	Talk + design activity	60 min	Classes 10–12
90	Glossary	Reference	—	All
91	Safety and Lab Conduct Guide	Reference	—	Teachers (required)
92	Quizzes and Answer Keys	Assessment	10–15 min per quiz	All

Times are suggestions based on a typical class; adjust them to your group.

2. Suggested pathways

Pathway	Order	What students gain
Core: from genes to cloning (3 sessions)	01 → 02 → 03 → 04	The flow of information from DNA to protein; how bacteria control genes; how scientists engineer them
Microbe lab days	05 → 06 → (07)	How bacteria are grown and studied; sterile technique; how viruses are detected
Environment module	11 → 12	Ecosystems, food webs, and how poisons build up in food chains
Chemistry of life workshop	08 → 09 → 10	Enzymes and metabolism in action; evidence and forensic science
Capstone	13 + 91	Design thinking: a biosensor and the responsibility that goes with it

HOW THE IDEAS CONNECT

- **DNA → RNA → protein** (01) is the basis of everything else.
- The **lac operon** (03) uses the ideas from 01 and leads to **blue–white screening** in 04.
- **PCR** (07) is built on **DNA replication** (01) and is used in **cloning** (04).
- **Enzymes** (08) explain DNA polymerase, restriction enzymes, ligase and β-galactosidase.
- The **culture** work (05–06) gives the practical skills behind **transformation** (04).

3. Materials checklist

Activity	Main materials	Prepare ahead
Codon Lottery (02)	Printed slips, codon chart, container	Print and cut 48 slips per team
Agar plates (06)	Premade nutrient agar plates, marker, tape, sanitiser, loops, flame source	Order plates early; check strain and incubator
Yeast balloons (09)	5 bottles, balloons, yeast, 4 sweeteners, string	Weigh sweeteners
Fingerprints (10)	Index cards, pencil, ink pad, tape; iodine kit for teacher	Safety check for iodine
Catalase demo (08)	Potato, 3 % hydrogen peroxide, cups	Boil and chill some potato
Biomagnification (12)	Sweets (30 per team, 10 of one colour), 14 cups	Count and sort sweets
Ecology game (11)	Pictures of three ecosystems	Print or project

4. Teaching approach

- **Ask before you tell.** Open with a question or puzzle (for example "Can DNA be round?") and let students guess.
- **Use pictures and analogies.** Genome = cookbook; gene = recipe; ribosome = kitchen; restriction enzymes = scissors; ligase = glue. Keep slides mostly diagrams, with very little text.
- **Little and often.** Alternate 10–15 minutes of explanation with 5–10 minutes of doing: a worksheet, a game or a demonstration.
- **Check understanding as you go.** Use the quick questions at the end of each document or the quizzes in document 92.
- **Let students think like scientists.** Predict, test, record, and explain results — including "wrong" ones.

Making sessions inclusive

- **Language:** summary versions are provided in Hindi and Bengali. Explain key English terms in the local language, and let students answer in the language they are most comfortable with.
- **Different starting points:** do not assume prior knowledge. Define words when you first use them (document 90 helps) and avoid jargon.
- **Group work:** rotate roles (reader, recorder, speaker, materials manager) so that every student takes a turn at each.
- **Role models:** include scientists with different backgrounds and genders — for example Rosalind Franklin (DNA structure) and Fanny Hesse (agar), who are in these notes.
- **Access:** read out text on slides, describe diagrams aloud, use large print, and provide extra time where needed.
- **Safe space:** welcome all questions; there are no silly ones.

Adapting for different ages

Group	Suggestions
Classes 6–8	Skip equations and enzyme names. Use the analogies, games (02, 11, 12), the yeast balloons (09) and simple agar plates (06).
Classes 9–10	Use the full talk with diagrams, the worksheets and the hands-on activities. Leave out the details of replication, splicing and the Ct-value maths.
Classes 11–12	Use everything, including the numbers (doubling, transformation efficiency, protein structure) and the design activity (13).

5. Assessment

- **Exit ticket:** each student writes one thing they learned and one question they still have.
- **Quizzes:** use the short quizzes (document 92), alone or as a team contest.

- **Lab books:** students record aim, method, results and a conclusion for each hands-on activity.
- **Teach-back:** ask pairs to explain a concept to another pair using a drawing.